# Unsteady 2-D Convection-Diffusion Thermal Analysis

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# 1 Method Description

The goal is to solve the unsteady two–dimensional convection–diffusion equation

$$\frac{\partial T}{\partial t} + u(y)\frac{\partial T}{\partial x} = \alpha \left( \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right) \tag{1}$$

on the domain

$$0 \leq x \leq 10, \qquad -1 \leq y \leq 1 \tag{2}$$

with thermal diffusivity

$$\alpha = 0.01. \tag{3}$$

The velocity field is assumed laminar and steady,

$$u(y) = u_{\max}(1 - y^2), \tag{4}$$

with

$$u_{\max} = 1.5. \tag{5}$$

## 1.1 Boundary Conditions

**Inlet at $x = 0$ (Dirichlet)**

$$T(0, y, t) = (1 - y^2)\left[1 + A\sin(2\pi f t)\right] \tag{6}$$

**Bottom wall at $y = -1$ (Dirichlet)**

$$T = 0 \tag{7}$$

**Top wall at $y = 1$ (Neumann)**

$$\frac{\partial T}{\partial y} = 0 \tag{8}$$

**Outlet at $x = L_x$ (Neumann)**

$$\frac{\partial T}{\partial x} = 0 \tag{9}$$

## 1.2 Initial Condition

$$T(x, y, 0) = 0 \tag{10}$$

### 1.3 Grid Discretization

A uniform computational grid is used,

$$x_i = i\,\Delta x, \qquad i = 0, 1, \ldots, N_x, \tag{11}$$
$$y_j = -1 + j\,\Delta y, \qquad j = 0, 1, \ldots, N_y, \tag{12}$$
$$t^n = n\,\Delta t, \qquad n = 0, 1, 2, \ldots \tag{13}$$

### 1.4 Crank–Nicolson Formulation

The Crank–Nicolson (trapezoidal) method produces a linear system

$$a_p T_p = \sum a_{\text{nbr}} T_{\text{nbr}}, \tag{14}$$

where the coefficients depend on

$$\Delta t, \quad \Delta x, \quad \Delta y, \quad u(y), \quad \alpha. \tag{15}$$

### 1.5 Iterative Solution

The system is solved using Successive Over–Relaxation (SOR).

The initial guess is

$$T_{ij}^{(0)} = T_{ij}^{n}. \tag{16}$$

Interior nodes are updated using a Gauss–Seidel sweep

$$T_{ij}^{(k+1)} = (1-\omega)T_{ij}^{(k)} + \omega T_{GS}. \tag{17}$$

Convergence is checked using

$$\left| T_{ij}^{(k+1)} - T_{ij}^{(k)} \right| < \text{tol}\,. \tag{18}$$

After convergence, the temperature field $T(x, y, t)$ is used to compute

$$Q(t), \qquad Q_{\text{bottom}}(t), \qquad Q_{\text{in}}(t), \qquad Q_{\text{out}}(t), \qquad \frac{dQ}{dt}.$$

## 2 Finite Difference Approximation of the PDE

Define the spatial operator

$$L(T) = -u(y)\frac{\partial T}{\partial x} + \alpha\left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2}\right). \tag{19}$$

## 2.1 Spatial Discretization

The first derivative in $x$ (convection) is approximated using a central difference

$$\frac{\partial T}{\partial x} \approx \frac{T^n_{i+1,j} - T^n_{i-1,j}}{2\Delta x}. \tag{20}$$

The second derivative in $x$ is

$$\frac{\partial^2 T}{\partial x^2} \approx \frac{T^n_{i+1,j} - 2T^n_{i,j} + T^n_{i-1,j}}{\Delta x^2}. \tag{21}$$

The second derivative in $y$ is

$$\frac{\partial^2 T}{\partial y^2} \approx \frac{T^n_{i,j+1} - 2T^n_{i,j} + T^n_{i,j-1}}{\Delta y^2}. \tag{22}$$

Substituting these into the discrete operator gives

$$L_h(T^n)_{ij} = -u_j \frac{T^n_{i+1,j} - T^n_{i-1,j}}{2\Delta x} + \alpha \left[ \frac{T^n_{i+1,j} - 2T^n_{i,j} + T^n_{i-1,j}}{\Delta x^2} + \frac{T^n_{i,j+1} - 2T^n_{i,j} + T^n_{i,j-1}}{\Delta y^2} \right]. \tag{23}$$

## 2.2 Crank–Nicolson Time Discretization

The time discretization is

$$\frac{T^{n+1}_{ij} - T^n_{ij}}{\Delta t} = \frac{1}{2} \left[ L_h(T^{n+1})_{ij} + L_h(T^n)_{ij} \right]. \tag{24}$$

This produces the linear system

$$A\,T^{n+1} = b. \tag{25}$$

# 3 Gauss–Seidel and Heat Finite Difference

The Gauss–Seidel estimate is

$$\overline{T}_p = \frac{b_p + b_w T_w + b_e T_e + b_s T_s + b_n T_n}{a_p}. \tag{26}$$

The SOR update is

$$T_p^{(k+1)} = (1-\omega) T_p^{(k)} + \omega \overline{T}_p. \tag{27}$$

### 3.1 Total Heat Content

$$Q(t) = \int_{-1}^{1} \int_{0}^{L_x} T(x, y, t)\, dx\, dy. \tag{28}$$

The discrete approximation is

$$Q(t) = \sum_{i=0}^{N_x} \sum_{j=0}^{N_y} T_{ij}(t)\, \Delta x\, \Delta y. \tag{29}$$

### 3.2 Heat Loss from the Bottom Wall

$$Q_{\text{bottom}}(t) = -\alpha \int_{0}^{L_x} \left. \frac{\partial T}{\partial y} \right|_{y=-1} dx. \tag{30}$$

The derivative is approximated using interior nodes

$$\left. \frac{\partial T}{\partial y} \right|_{y=-1} = \frac{T_{i,2} - T_{i,1}}{\Delta y}. \tag{31}$$

The resulting discrete form is

$$Q_{\text{bottom}}(t) \approx -\alpha \sum_{i=0}^{N_x} \frac{T_{i,2} - T_{i,1}}{\Delta y} \Delta x. \tag{32}$$

### 3.3 Energy Balance Check

$$\frac{dQ}{dt} \approx \frac{Q(t_n) - Q(t_{n-1})}{\Delta t}. \tag{33}$$

## 4 Results

### 4.1 Global Heat Content vs Time

Global heat content $Q(t)$ was plotted for both the steady inlet case ($A = 0$) and the sinusoidal inlet case ($A = 0.5, f = 1$ Hz). In both cases $Q(t)$ grows approximately linearly at early times and then the slope decreases as heat begins to exit through the outlet and the bottom wall.

### 4.2 Bottom Wall Heat Loss vs Time

Bottom-wall heat loss $Q_{\text{bottom}}(t)$ was also plotted for $A = 0$ and $A = 0.5$. In both cases the curve starts near zero and becomes increasingly negative in time, indicating that the cold wall at $T = 0$ absorbs more heat as the fluid warms. The sinusoidal inlet case produces almost the same trend, with very small modulation.

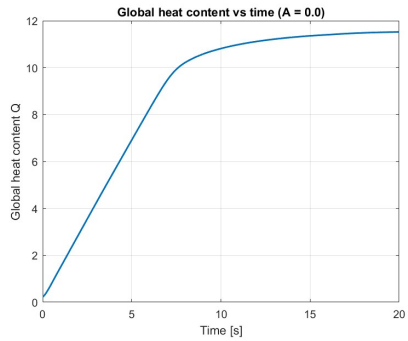


Figure 1: Global heat content vs time. No sinusoidal input (A=0) and 20 second run time. Mesh size chosen was Nx=50, Ny=40, and dt=0.02

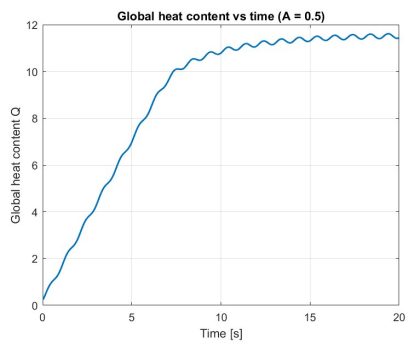


Figure 2: Global heat content vs time. Includes sinusoidal input (A=0.5 and f=1) and 20 second run time. Mesh size chosen was Nx=50, Ny=40, and dt=0.02

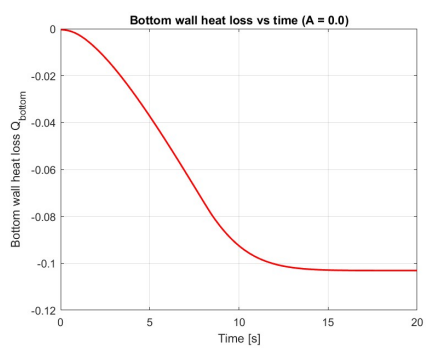


Figure 3: Bottom wall heat loss (A=0)

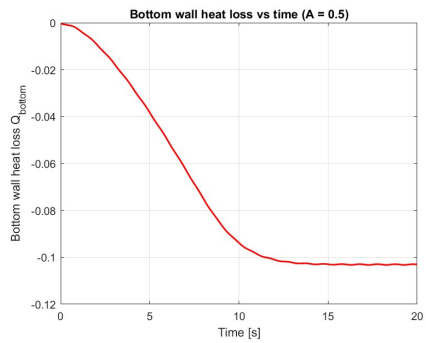


Figure 4: Bottom wall heat loss (A=0.5)

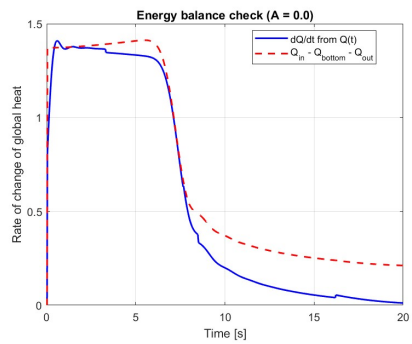


Figure 5: Energy balance (A=0)

## 4.3 Energy Balance vs Time

To visualize the energy balance over time, plots were generated comparing

$$\frac{dQ}{dt}$$

computed from the discrete $Q(t)$ and the computed quantity

$$Q_{\text{in}} - Q_{\text{bottom}} - Q_{\text{out}}$$

for both the steady and sinusoidal inlet cases.

For $A = 0$ the two curves agree well for early times (roughly 0–6 s) but deviate more as strong gradients develop near the outlet. For $A = 0.5$ the curves show clear oscillations at 1 Hz but remain close to each other over most of the simulation, indicating consistent energy accounting.

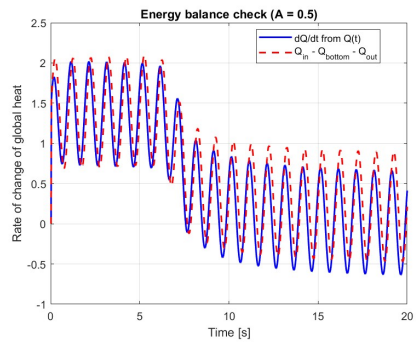


Figure 6: Energy balance (A=0.5)

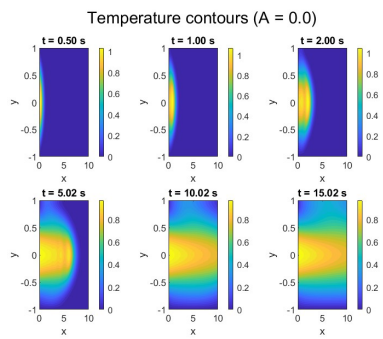


Figure 7: Temperature contours A=0

### 4.4 Temperature Contours and Animation

Temperature contour plots were generated at regular time intervals (every 0.4 s) by storing snapshots of $T(x, y, t)$ during the time integration. These contours show the growth and downstream convection of the heated region with time.

An animation was also created using matplotlib's `FuncAnimation`, showing the evolution of the temperature field over the whole simulation time for both $A = 0$ and $A = 0.5$. The animations visually confirm that the heated region grows, advects, diffuses, and interacts with the walls as expected.

## 5 Convergence Study

To study convergence, three grid/time-step combinations were tested:

- Coarse mesh: $N_x = 25$, $N_y = 20$, $\Delta t = 0.04$
- Medium mesh: $N_x = 50$, $N_y = 40$, $\Delta t = 0.02$
- Fine mesh: $N_x = 100$, $N_y = 80$, $\Delta t = 0.01$

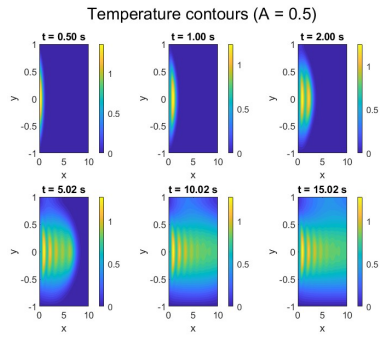


Figure 8: Temperature contours A=0.5, f = 1

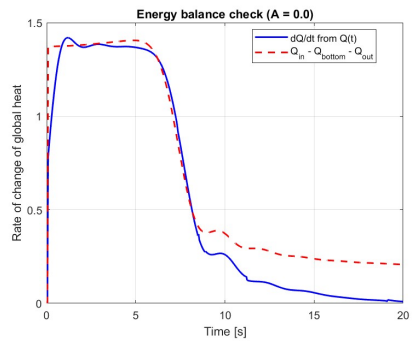


Figure 9: Energy balance coarse mesh

Because it would be overkill to analyze all graphs, the energy balance check was used for comparison between meshes. The other graphs showed almost no change.

Three observations were made. First, the fine grid mesh required significantly more computation time. Each factor of two increase in mesh resolution results in approximately eight times more calculations. Therefore the fine mesh required roughly sixty-four times as many calculations as the coarse mesh.

Second, all meshes produced nearly identical behavior. The bottom-wall heat and global heat plots were also very similar across meshes, suggesting that even the coarse mesh can be considered converged.

Third, the calculated derivative approaches the estimated value fastest for the smallest time step, indicating that although the spatial meshes are converged, the time step still influences how quickly the two energy terms match.

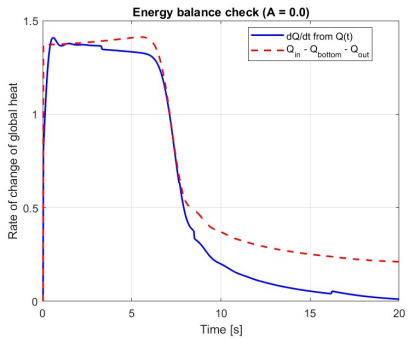


Figure 10: Energy balance medium mesh

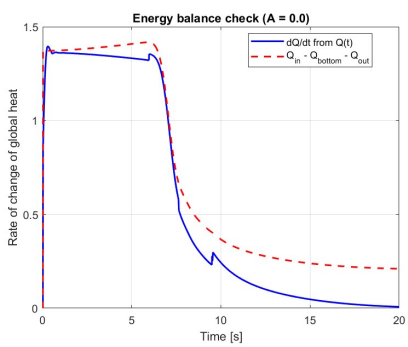


Figure 11: Energy balance medium mesh

# 6 Stability Analysis

Because Crank–Nicolson is equivalent to applying the trapezoidal rule to the semi-discrete ODE system

$$\frac{dT}{dt} = LT + s(t),$$

the scheme is $A$-stable for the diffusion part of the operator and is unconditionally stable with respect to $\Delta t$.

The addition of convection introduces imaginary components to the eigenvalues, leading to oscillatory but bounded behavior in time.

## 6.1 Positivity of Coefficients

The linear system can be written as

$$a_p T_{i,j}^{n+1} = \sum a_{\text{nbr}} T_{\text{nbr}} + b_p.$$

Previously we obtained

$$a_p = 1 + r_x + r_y > 0$$

$$a_w = \frac{r_x + c_j}{2} > 0$$

$$a_e = \frac{r_x - c_j}{2} > 0$$

which implies

$$r_x > |c_j|.$$

Substituting the definitions of the coefficients gives

$$\Delta x < \frac{2\alpha}{u_j}.$$

Using the worst-case velocity produces the bound

$$\Delta x < \frac{2\alpha}{u_{\max}} = 0.0133.$$

This value is much smaller than the grid spacing used in the simulations. Thus strict positivity is violated, but the scheme remains stable due to the unconditional stability of Crank–Nicolson.

Numerically it was observed that a relaxation factor $\omega = 1.7$ caused NaNs early in the simulation when gradients were largest, while $\omega = 1.3$ produced stable iterations.

# 7 Accuracy Analysis

All spatial derivatives use central differencing.

Using a Taylor expansion about $(x_i, y_j)$:

## 7.1 Convection Term

$$\frac{T^n_{i+1,j} - T^n_{i-1,j}}{2\Delta x} = T_x + \frac{\Delta x^2}{6} T_{xxx} + O(\Delta x^4)$$

so the truncation error is

$$O(\Delta x^2).$$

## 7.2 Diffusion Term

$$\frac{T^n_{i+1,j} - 2T^n_{i,j} + T^n_{i-1,j}}{\Delta x^2} = T_{xx} + \frac{\Delta x^2}{12} T_{xxxx} + O(\Delta x^4)$$

which is also

$$O(\Delta x^2).$$

The same analysis in the $y$ direction gives

$$O(\Delta y^2).$$

Crank–Nicolson provides second-order accuracy in time

$$O(\Delta t^2).$$

Therefore the overall truncation error is

$$O(\Delta x^2) + O(\Delta y^2) + O(\Delta t^2).$$

# 8 Sinusoidal Inlet Effects and Graph Discussion

To study the effect of an unsteady inlet, simulations were repeated with sinusoidal amplitude $A = 0.5$ and frequency $f = 1$ Hz.

## 8.1 Effect on Global Heat Content

For $A = 0$, $Q(t)$ increases smoothly and almost linearly until roughly $t \approx 7$ s. After that point the slope decreases as the heated region approaches the outlet and heat loss increases.

For $A = 0.5$ the same overall trend appears but with small oscillations at 1 Hz. Even though the inlet temperature oscillates by $\pm 50\%$, the global heat shows only small oscillations because the domain averages out the inlet variations through convection and diffusion.

### 8.2 Effect on Energy Balance

For $A = 0$ the curves for $dQ/dt$ and $Q_{\text{in}} - Q_{\text{bottom}} - Q_{\text{out}}$ match well for early times but deviate as strong gradients reach the outlet.

For $A = 0.5$ both curves oscillate at 1 Hz and remain closely aligned for most of the simulation.

### 8.3 Effect on Bottom Wall Heat Loss

For both inlet cases the bottom-wall heat loss becomes more negative over time. The sinusoidal inlet has little effect because the cold wall acts as a strong heat sink and the oscillatory signal is diffused before reaching the wall.

### 8.4 Effect on Temperature Contours

For $A = 0$ the contours show smooth growth of the heated region. For $A = 0.5$ oscillations appear near the inlet but gradually disappear downstream due to diffusion and convection.

## 9 Solution Plausibility

Overall, the numerical results exhibit the expected behavior for unsteady convection–diffusion in a duct with a heated inlet and a cooled wall. The global thermal energy content increases approximately linearly at early times, as hot fluid enters the domain while comparatively little heat is removed. When a sinusoidal temperature (or heat-flux) condition is imposed at the inlet, temporal oscillations emerge in the solution at the prescribed forcing frequency. The heat-loss profile at the bottom wall becomes increasingly negative, indicating a net heat uptake by the wall from the progressively warming fluid. Energy-balance diagnostics demonstrate close agreement between $dQ/dt$ and $Q_{\text{in}} - Q_{\text{bottom}} - Q_{\text{out}}$, particularly in the initial stages of the simulation. Collectively, these results suggest that the computed solution is physically consistent and that the numerical scheme exhibits stable and reliable behavior.